Automated Time-Lapse Microscopy — Inženernoe reshenie + budget

Цель: Round-the-clock наблюдение за BJ-hTERT fibroblast cultures в Phase A CDATA эксперименте. Автоматическое сканирование, auto-focus, периодический imaging (brightfield + fluorescence), remote monitoring.

Базовая платформа: Zeiss IM 35 inverted microscope (существующий)

Функциональные требования

1. **Automated XY movement** — 96-well plate или multi-dish scanning (5-10 positions per session)
2. **Automated Z-focus** — autofocus на каждой position для long-term drift compensation
3. **Time-lapse scheduling** — imaging every 30-60 min, 24/7, до 3 недель непрерывно
4. **Brightfield + fluorescence imaging** — GT335/polyE (FITC) + Ninein (TRITC), caught in same session
5. **Live cell environment** — 37°C, 5% CO_2, humidity control, sterile
6. **Remote monitoring** — Jaba может смотреть из дома без физического присутствия
7. **Data storage & auto-processing** — NAS или cloud backup, ImageJ/Fiji batch pipeline
8. **Power redundancy** — UPS (минимум 2 часа при blackout)
9. **Alert system** — SMS/email если что-то нарушилось (temperature drift, CO_2 loss, camera crash)

Инженерное решение — 3 варианта

Вариант A — Entry-level retrofit ($3,500–5,000) RECOMMENDED для Phase A

Philosophy: DIY + open-source. Максимально дёшево, достаточно для 6-месячного Phase A.

Компонент	Модель / источник	Цена
Motorized XY stage adapter	Arduino CNC-shield + stepper motors + linear rails (custom) or OpenFlexure-style DIY kit	$500
Motorized focus drive	NEMA-17 stepper + belt drive on Zeiss coarse knob (3D-printed mount)	$200
Digital camera	FLIR Blackfly S USB3 BFS-U3-63S4M-C (mono, 2448×2048, scientific)	$1,200
Live cell chamber	OkoLab bold-line стандарт ($1,500) OR DIY acrylic chamber + Peltier heater + CO_2 solenoid ($400)	$400 (DIY) or $1,500 (pro)
CO_2 + temperature controller	OkoLab H301 or Tokai Hit INU compact ($800) OR DIY with CO_2 tank + Arduino + DHT22 sensor + solenoid ($250)	$250 (DIY)
LED fluorescence illuminator	SOLA SE II Lumencor ($1,500 used) or ThorLabs M470L4 + M565L3 + filter cube ($500)	$500 (LED stand-alone)
Computer	Refurbished Dell OptiPlex i5 16GB RAM 512GB SSD ($300)	$300
Acquisition software	**Micro-Manager 2.0** (free, open source) + **PyMMCore-Plus** Python bindings	$0
Data processing	**Fiji** (free) + custom Python pipeline	$0
UPS	APC Back-UPS Pro 1500VA ($200)	$200
Remote monitoring	VPN (WireGuard free) +	$0

Компонент	Модель / источник	Цена
	VNC/RustDesk (free)	
NAS backup	Synology DS220+ with 2×4TB HDD ($500)	$500
Misc (cables, mounts, 3D prints)		$150
Подитог Вариант A		**$4,200**

Плюсы: Влезает в Phase A $80k с запасом. Полный control. Open-source. **Минусы:** DIY = 2-3 недели setup time. Зависит от personal engineering skills. Calibration может дрейфовать.

Вариант B — Mid-range retrofit ($8,000–12,000)

Philosophy: Quality off-the-shelf компоненты, меньше DIY.

Компонент	Модель	Цена
Prior 3rd-party motorized XY stage	Prior ProScan III H117 с controller	$3,500 (used eBay)
Prior motorized focus (Z)	Prior Z-Drive motor + adapter	$1,500 (used)
Digital camera	Hamamatsu ORCA-Fusion BT ($6k new, $2k used)	$2,000 (used)
Live cell chamber + CO_2	OkoLab H301-K-Frame bold-line complete	$2,500 (used)
LED fluorescence	Lumencor SOLA SE II	$1,500 (used)
Computer	Dell Precision 3650 Xeon, 32GB RAM, 1TB NVMe	$800
Software	Micro-Manager (free) or µManager Studio	$0
UPS + NAS + miscellaneous		$900
Подитог Вариант B		**$12,700**

Плюсы: Industrial-grade reliability. 2-3 days setup. No calibration issues. **Минусы:** Дороже. Больше overhead для Phase A bootstrap.

Вариант C — Complete turnkey ($25,000–50,000)

Philosophy: Buy ready-to-use automated microscopy system, ignore Zeiss IM 35.

- **Nikon Eclipse Ti2-E** motorized inverted + Perfect Focus System (PFS) + DS-Qi2 camera + NIS-Elements HCA-module + OkoLab environmental
- Used market price: $25,000–50,000 depending on age and configuration

Плюсы: State-of-the-art, plug-and-play. **Минусы:** Выходит за budget Phase A; может быть revisited для Phase B или post-grant.

Recommended для LOI v24: Вариант A ($4,200)

Укладывается в существующую строку бюджета "Microscope Upgrade ($2,000)" — нужно **increase to $4,500** (с reserve $300) и перераспределить:

Новая budget line в LOI Phase A

Раздел "Microscope Upgrade & Automation": $4,500 (было $2,000)

Sub-item	Cost
Motorized XY stage (DIY kit / Arduino CNC)	$500
Motorized Z focus drive	$200
FLIR Blackfly S digital camera	$1,200
LED fluorescence illuminator (ThorLabs M470L4+M565L3)	$500
CO_2 + environmental DIY controller	$250
DIY live cell chamber (acrylic + heater)	$400
Computer (refurbished)	$300
UPS 1500VA	$200
NAS backup system	$500
3D-printed mounts + cables + misc	$150
Setup contingency	$300
Total	**$4,500**

Где взять +$2,500 (relative to existing LOI):

- Снизить "General Consumables" с $12k на $10k (тщательный бюджет)
- Либо снизить "Technician Salary" с $18k на $17k (1 мес меньше FTE)
- Либо добавить строку "Automation Equipment" = $4,500 отдельно и увеличить total request на $2,500 → $82,500 (legal в Impetus, limit $150k per phase)

Timeline for setup

Week	Task
1	Ordering components (Arduino, FLIR, chamber, LED, NAS)
2	Delivery + initial assembly
3	Motorized stage integration + Z-focus motor mounting
4	Camera + Micro-Manager installation + calibration
5	Environmental chamber + CO_2 flow testing
6	LED fluorescence setup + filter alignment
7	Full system integration test (dry run without cells)
8	First cell experiment (BJ-hTERT validation)
9-24	Phase A data collection (experiments Aim A.1, A.2, A.3)

Setup occupies Months 1-2 of Phase A budget. Experiments begin Month 3. Data analysis through Month 6.

Technical specifications

Imaging parameters

Parameter	Value
Resolution	2448×2048 px (FLIR BFS-U3-63S4M)
Bit depth	12-bit → 16-bit stored
Frame rate	74 FPS max (not limiting for time-lapse)
Exposure	50-500 ms (brightfield), 500-3000 ms (fluorescence)
Z-stack	5-11 slices × 2 μm spacing (optional for 3D imaging)
Channels	Brightfield + FITC (GT335/polyE) + TRITC (Ninein)
Time-lapse interval	30 min (day/night)
Data per session	~5 GB/day (if 10 positions × 3 channels × 48 timepoints)
Total data 6 mo	~900 GB (easily fits NAS 4TB)

Stage automation

- **XY range:** Ensure coverage of 35mm dish (~40mm × 40mm) or 6-well plate
- **XY accuracy:** ±5 μm positioning (achievable with Arduino CNC-style steppers + linear rails; Prior stages are ±0.1 μm but more expensive)
- **Z range:** 10mm travel, ±0.5 μm accuracy (adequate for autofocus)
- **Autofocus:** Software-based, contrast detection on DAPI/Hoechst channel OR hardware laser-based (more expensive, Перспектива в upgrade)

Environmental control

- **Temperature:** 37°C ±0.3°C (Peltier + PID controller)
- **CO_2:** 5% ±0.2% (mass flow controller + CO_2 tank; one tank lasts ~2 weeks)
- **Humidity:** 95% (passive evaporation from humidifier insert)

Remote monitoring setup

1. **VPN** (WireGuard) into lab network from Jaba's home
2. **Micro-Manager live display** via VNC/RustDesk
3. **Camera preview stream** via MJPEG over HTTP (for quick peek)
4. **SMS alerts** via Twilio API ($0.01/msg, ~$1/month) when:
    - Temperature deviates >0.5°C
    - CO_2 deviates >0.5%
    - Camera disconnects
    - UPS engaged (power failure)
5. **Daily email summary** — automated Python script sends overview image + environmental log at 8am

---

Список для закупки (ordering list)

Arduino / electronics (AliExpress / RobotDyn / Amazon — $500 total)

- Arduino Mega 2560 R3 ($25)

- CNC Shield V3 ($10)
- 3× NEMA-17 stepper motors (X, Y, Z) ($40)
- 3× A4988 stepper drivers (already on shield)
- Linear rails 400mm × 2 + 200mm × 2 (X, Y) with bearings + belt drive ($120)
- 12V 10A power supply ($30)
- Endstops × 6 ($10)
- Wires + breadboard + misc ($30)
- DHT22 temperature/humidity sensor ($10)
- CO_2 solenoid valve 12V ($40)
- MH-Z19B CO_2 sensor ($25)
- DS18B20 waterproof temp probe ($10)
- MOSFET modules × 4 ($20)
- Peltier heater module + heatsink ($40)
- 3D-printed parts (stage adapter, focus adapter, chamber mount) — print yourself or Shapeways ($90)

Camera + optics ($1,700 total)

- FLIR Blackfly S USB3 BFS-U3-63S4M-C monochrome scientific camera ($1,200, new)
- C-mount to Zeiss IM 35 adapter ($80)
- ThorLabs M470L4 blue LED (470 nm, for FITC excitation) ($180)
- ThorLabs M565L3 green LED (565 nm, for TRITC excitation) ($180)
- ThorLabs LEDD1B LED driver ($60)

Live cell environmental ($1,050 total)

- Acrylic chamber (custom-cut 4mm thick, ~200mm × 200mm × 60mm internal, $120)
- Glass cover slip for bottom ($20)
- CO_2 tank 9kg (refillable, $60 Tbilisi medical supply)

- CO_2 regulator + flow meter ($80)
- Peltier element TEC1-12706 + heatsink + fan ($25)
- PID controller (Inkbird ITC-100 or equivalent) ($35)
- Humidifier insert (plastic tray with foam) ($10)
- OKolab compact CO_2 incubator mini chamber (used market, optional upgrade, $700)

Computer + storage ($1,000 total)

- Refurbished Dell OptiPlex (i5-8500, 16GB, 512GB SSD, Win10 Pro) ($300)
- Synology DS220+ NAS ($230) + 2× 4TB Seagate IronWolf ($180) = $590
- APC Back-UPS Pro 1500VA ($200)
- External monitor 24” (existing or $150)

Total Variant A: $4,250 + shipping/VAT (~$200) = **$4,450**

LOI v24 budget amendment

Текущая Phase A budget: $80,000

Предлагаемое изменение:

- Уменьшить “General Consumables” с $12,000 на $9,500 (-$2,500)
- Увеличить “Microscope Upgrade” с $2,000 на $4,500 (+$2,500)
- Переименовать line в “**Microscope Automation & Upgrade**”
- Добавить footnote: “Includes motorized XY + Z stage for round-the-clock imaging, FLIR digital camera, LED fluorescence illumination, environmental chamber (37°C + 5% CO_2), NAS backup, and UPS. Full specification in supplementary materials.”

Net effect: $80,000 total unchanged. Более точная allocation для automation.

⚠ Key risks & mitigations

1. **DIY stage calibration drift** → use linear rails с encoders (add ~$100); alignment every 2 weeks

2. **CO_2 tank run-out mid-experiment** → install low-pressure sensor + SMS alert; keep 2 tanks rotating
3. **Power outage > UPS runtime** → UPS gives 2h; manually restart if needed; alert sent immediately
4. **Contamination** (long-term cultures) → weekly media change, sterile technique, monthly bleach decontamination of chamber
5. **Camera failure** → budget $500 contingency for replacement (not in main budget; covered by $300 setup contingency + technician hours)
6. **Software crash** → Micro-Manager + watchdog script: if no new frame in 2h, restart acquisition automatically
7. **Remote access compromised** → WireGuard with ed25519 keys; no public SSH; lab IP whitelist

---

Claude Code /overnight — AI-operated microscopy

Концепция

Microscope управляется **Claude Code** (Anthropic's official CLI для AI-assisted workflows) в специальном **/overnight режиме** — автономный цикл наблюдения, принятия решений и корректировки эксперимента ночью, когда человек не в лаборатории (20:00–08:00).

В /overnight режиме agent получает авторизацию на **routine decisions** (выбор поле зрения, autofocus, channel switching, ROI refinement) без ежеминутного одобрения человека. **Strategic decisions** (изменение протокола, остановка эксперимента) остаются у человека.

Архитектура

```
┌──────────────────────────────────────────────────────────┐
│            HUMAN (Jaba, 20:00–08:00 off-duty)            │
│                 ↕ SMS/email alerts only                  │
└──────────────────────────────────────────────────────────┘
                              ↕
┌──────────────────────────────────────────────────────────┐
│             CLAUDE CODE — /overnight agent               │
│  • Reads experiment protocol (PROMPT.md, см. ниже)       │
│  • Monitors image stream в real-time                     │
│  • Calls tool functions: move_stage(), autofocus(),      │
│      capture_multichannel(), switch_led(), etc.          │
```

```
│   • Decision loop: каждые 30 мин inspect last batch,        │
│     выбрать next action (stay/move/rotate/alarm)            │
│   • Journaled decisions с rationale                         │
│   • On blocker: pauses + SMS to experimenter                │
└─────────────────────────────────────────────────────────────┘
                         ↕ Python tool functions (PyMMCore-Plus)
┌─────────────────────────────────────────────────────────────┐
│  MICRO-MANAGER 2.0 core + pymmcore-plus + Arduino firmware   │
└─────────────────────────────────────────────────────────────┘
                                                               ↕
┌─────────────────────────────────────────────────────────────┐
│   Zeiss IM 35 + motorized XY/Z + camera + chamber + LEDs     │
└─────────────────────────────────────────────────────────────┘
```

Tool functions для Claude Code

```
# микроскоп-управление
move_stage_xy(x_mm, y_mm); move_stage_relative(dx, dy)
autofocus(mode='contrast'|'laplacian')
capture_single(channel, exposure_ms); capture_zstack(ch, z_range, step)
capture_multichannel(channels)
switch_led(channel, intensity_pct)

# image analysis
detect_cells(image); detect_centrioles(image)
measure_polyglu_intensity(cell_roi, channel)
segment_nuclei(image); count_dead_cells(image)

# environmental
get_temperature_c(); get_co2_percent(); get_humidity_percent()
set_target_temp(c); open_co2_valve(duration_s)

# alerting / journaling
send_sms_to_human(message, priority='info'|'warn'|'crit')
log_decision(action, rationale, observed)
save_checkpoint(state); halt_experiment(reason)  # requires human unlock
```

Cost

Claude Code subscription: **$20/мес Pro tier** × 6 мес = **$120 total** — negligible vs $4,500 hardware.

PROMPT-based experimentor signaling

Концепция

Experimentator (Jaba) пишет **PROMPT** на естественном языке описывающий **цели и задачи эксперимента**. Claude Code интерпретирует этот prompt и автоматически:

1. **Мониторит данные** в реальном времени (image stream, environmental log)
2. **Проверяет цели** каждые 30 минут
3. **Отправляет сигналы** человеку когда происходит что-то важное: достижение цели, anomaly, риск для эксперимента, необходимость intervention

Преимущество: Jaba не нужно программировать условия срабатывания. Он описывает цели словами — Claude сам формулирует критерии и мониторит.

Формат PROMPT файла

~/Documents/Experiments/CDATA_PhaseA_Aim1/PROMPT.md:

Aim A.1 — α vs β discrimination

Цель

Disentangle division-dependent (α) from time-dependent (β) polyGlu accumulation in BJ-hTERT fibroblasts by comparing proliferating vs contact-inhibited conditions.

Условия

- **Proliferating wells** (A1-A6): sub-confluent, passaged every 3 days
- **Contact-inhibited wells** (B1-B6): seeded confluent, media changed 2× per week
- **Time points for imaging**: days 0, 7, 14, 21
- **Channels**: Brightfield, FITC (GT335 polyGlu), TRITC (Ninein mother centriole)

Что измеряем

- polyGlu intensity at mother centriole (FITC co-localized with TRITC)
- cell density (brightfield segmentation)
- morphology changes (senescence markers: flattened cells, vacuoles)

Сигналы, которые я хочу получить

Позитивные (progress toward goal)

- **"Signal detected"**: если polyGlu intensity в proliferating wells вырос на >30% vs day 0 baseline
- **"$\alpha > 0$ confirmed"**: если при day 14 in proliferating wells средний intensity significantly higher ($p<0.05$, t-test) vs contact-inhibited wells
- **"Parrinello signal"**: если 20% O_2 wells показывают >40% difference vs 3% O_2 wells (for O_2 modulation Aim A.2)

Предупреждения (possible trouble)

- **"Contamination risk"**: если cell density drops >20% unexpectedly в 2+ wells
- **"Focus drift"**: если autofocus fails в 3+ positions consecutively
- **"Channel saturation"**: если FITC mean intensity > 60,000 counts (max 65,535)
- **"Morphology anomaly"**: если >15% cells show senescent markers earlier than expected

Критические (требуют немедленного вмешательства)

- **"Experiment compromised"**: если cell death >40% in any well → stop imaging, SMS Jaba, save last data
- **"Environment failure"**: если temp <35°C, CO_2 <4% или >6%, humidity <80% — pause, SMS Jaba
- **"Hardware fault"**: если camera disconnects, stage jams, UPS engaged — halt, SMS Jaba

Что Claude может делать автономно

- Выбирать best focal plane для imaging
- Adjust LED intensity within ±20% if signal too dim/bright
- Re-image if blur detected
- Skip dead cells в analysis
- Increase imaging frequency на interesting positions (if Claude detects cell division event)
- Request Z-stack если 2D image недостаточно

Что Claude НЕ делает без моего разрешения

- Менять protocol (пример: не добавляет лишние channels)
- Остановить experiment раньше scheduled end
- Открыть chamber (sterility)
- Изменить cells физически (no chemical addition)

Формат отчётов

- **Daily 8 AM email**: summary of last 24 hrs + 4 sample images
- **Mid-day SMS (14:00)**: short status (OK / warning / critical)
- **Cycle log (every 30 min)**: в JOURNAL.md, no notification to human unless signaled above
- **End-of-experiment report**: day 21, full analysis, recommended next steps

Контекст для Claude (для понимания важности сигналов)

Данный эксперимент — Aim A.1 of Impetus Phase A ($80k grant). Результат определяет
go/no-go decision для Phase B ($120k). CDATA hypothesis предсказывает $\alpha > 0$ значимо.
Null result ($\alpha \approx 0$ и $\beta \approx 0$) = xonest falsification, publish negative result.

Как Claude использует PROMPT

1. **Парсинг целей:** извлекает из текста конкретные метрики (polyGlu intensity, cell density, effect sizes) и thresholds (>30%, >40%, $p<0.05$)
2. **Постоянная проверка:** каждые 30 мин на основе последних imaging data проверяет — сработало ли какое-то условие из “Сигналы”
3. **Генерация сигнала:**
    - “Signal detected” → send_sms с кратким summary и attached preview image
    - “Contamination risk” → send_sms + предлагает actions (“recommend: check wells A3, A5”)
    - “Experiment compromised” → halt + SMS + save checkpoint
4. **Log of reasoning:** каждый decision журналируется в JOURNAL.md с quoting the relevant prompt line и показателями

Пример SMS сигналов

Scenario 1 — Positive progress:

[CDATA-A1] 14:30 | INFO
$\alpha > 0$ confirmed: proliferating wells show 2.3x higher polyGlu vs contact-inhibited at day 14 ($p=0.003$, $n=6$ wells each). Keeping protocol. Full data: dashboard/cycle_156

Scenario 2 — Warning:

[CDATA-A1] 03:15 | WARN
Cell density dropped 28% in wells A4, A5 between 02:00 and 03:00 imaging. Possible contamination. Recommend visual check at 8 AM. Wells A1-A3, A6 normal. Continuing schedule, flagged for review.

Scenario 3 — Critical:

[CDATA-A1] 05:42 | CRIT
CO_2 dropped to 3.1% (target 5.0%). Opened valve 45s × 3, no recovery. Tank may be empty. HALT imaging. Cells survive ~4 hrs at low CO_2. Action needed before 10 AM. Checkpoint saved.

Как adaptировать PROMPT для разных aim'ов

Aim A.2 (20% vs 3% O_2): добавить signals для differential wells; modify critical thresholds для O_2-sensitive markers.

Aim A.3 (CCP1/TTLL6-OE causal): добавить signals for population doubling rates, transduction efficiency (GFP% cells), senescence onset timing.

Phase B (HSC mouse): полностью другой PROMPT.md — focus на chimerism, flow cytometry results, не microscopy.

Template для experimentor

```
# [Aim Name]

## Цель

[один параграф о scientific goal]

## Условия

[список conditions: wells, channels, time points]

## Что измеряем
```

[metrics to quantify]

Сигналы

Позитивные (progress)

- [condition] → signal "[name]": [action]

Предупреждения

- [condition] → signal "[name]": [action]

Критические

- [condition] → signal "[name]": HALT + SMS Jaba

Автономные действия Claude

[что можно без authorization]

НЕ делать без разрешения

[что требует human approval]

Отчёты

- [частота: daily/mid-day/end-of-experiment]
- [формат: email/SMS/dashboard]

Контекст

[почему это важно, stakes, acceptable null result]

Преимущества prompt-driven approach

1. **Zero programming:** Jaba описывает словами, не пишет Python
2. **Domain-expert friendly:** biologist формулирует условия в биологических терминах, Claude переводит в thresholds
3. **Easy iteration:** менять цели эксперимента = редактировать PROMPT.md, без code changes

4. **Explanatory:** Claude’s decisions всегда ссылаются на конкретные lines PROMPT.md — transparency
5. **Reusable:** PROMPT.md template для каждого Aim, можно делиться с коллегами
6. **Fail-safe:** если PROMPT не покрывает сценарий — Claude не действует autonomously, pause и запрашивает human

Validation перед submission

Перед тем как Claude запускает /overnight, Jaba делает:

1. **Dry run** на синтетических данных — проверить что сигналы срабатывают корректно
2. **Day-1 validation**: первые 24 часа Claude в /overnight режиме, но Jaba проверяет logs каждый час вручную
3. **Progressive автономность:** после 48 часов без issues — full autonomy; если issue — уменьшить autonomy

Для Impetus LOI v24 — Methodological Innovation

Этот prompt-driven signaling — дополнительный **scientific selling point**:

> “Our automated microscopy platform pairs low-cost retrofit ($4,500) with **prompt-driven AI supervision** (Claude Code /overnight). The experimenter articulates goals и thresholds в natural-language PROMPT.md; Claude continuously monitors data and signals the human only when strategically important events occur (positive progress, warnings, critical failures). This eliminates the ‘experimenter wake-up cost’ traditional time-lapse requires and democratizes AI-assisted experimental science for resource-limited labs. All PROMPT templates, tool functions, and decision logs released under MIT license concurrent with Phase A preprint.”

Follow-up upgrades (post-Phase A, if Go decision)

If Phase A passes и funding extends (Phase B or follow-on grant):

- **Hamamatsu ORCA-Fusion BT** upgrade (+$4k used) → higher sensitivity for low-expression fluorescence
- **Perfect Focus System** (Nikon PFS) retrofit (+$3k used) → hardware-based autofocus, no software drift

- **Motorized filter turret** (+$1k) → automated channel switching, faster imaging
- **Liquid handling robot** (OpenTrons OT-2 $2.5k new) → automated media change, ideal for 3-4 week contact-inhibition protocols

Budget for upgrades: place into Phase B or post-grant $10-15k capital line.

Decision for LOI v24 submission (2026-04-24)

Recommendation: Вариант A ($4,500 total equipment line) — максимально cost-effective, polностью покрывает Phase A experimental requirements (automated day-night imaging, environmental control, remote monitoring, data backup). Upgradeable to Phase B levels later.

Action:

1. Amend Phase A budget: “Microscope Upgrade” $2k → “Microscope Automation & Upgrade” $4.5k (reallocate $2.5k from Consumables)
2. Add this document as supplementary material
3. Include in Cover Letter: “A custom-engineered automated time-lapse imaging platform (see Supplementary: Automated Microscopy Setup) will enable round-the-clock tracking of centriolar polyGlu dynamics in live fibroblasts, maximizing temporal resolution within budget constraints.”

Engineered 2026-04-21 for Impetus LOI v24 submission 2026-04-25.